Paradigms of Machine Learning

1

Pre

CS 3244
Machine Learning

NUS National University of Singapore | Computing

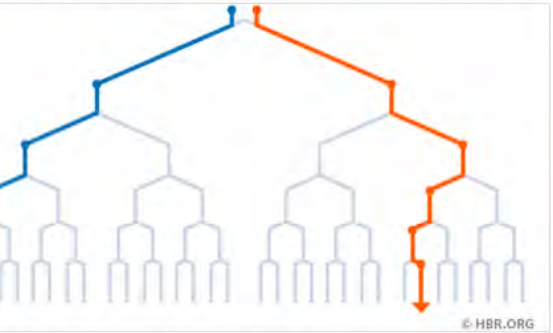
fference Between AI, Machine Learning, and
earning

ne learning, and deep learning are terms that are often used
geably. But they are not the same things.

IDIA.COM

Harvard Business Review · Like Page · Sign Up
October 22, 2015 ·

learning, explained.



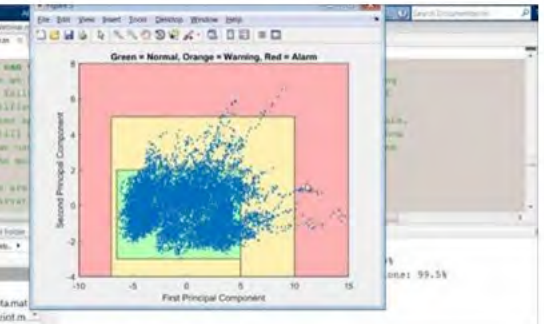
Machines Learn (And You Win)

smart predictions with statistical models

G

ATLAB · Like Page · Message
19 at 10:55pm ·

undamentals of machine learning with MATLAB in this video!



ne Learning Made Easy

Get ready for this year's AI-centric EmTech Digital, starting with our Q+A with the leader of Google Brain.

Machine Learning for Everyone

Recent advances are making machine learning useful outside the tech industry, says the leader of the Google Brain research group.

TECHNOLOGYREVIEW.COM

World Economic Forum · Like Page · Sign Up
June 18 ·

"You may not know it, but it's all around you."

Why you need to understand machine learning

From picking stocks to finding partners, machine learning plays a part in every stage of our lives - yet most of us know little about these powerful tools.

WEFORUM.ORG

The Economist · Like Page · Sign Up
May 14, 2015 ·

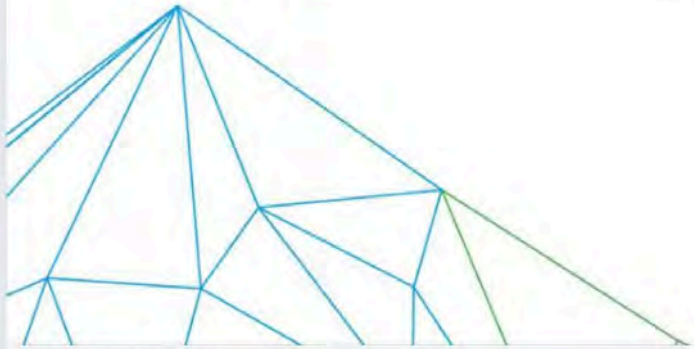
The standard joke about artificial intelligence (AI) is that, like nuclear fusion, it has been the future for more than half a century now. AI is in the news again, for there has been impressive progress in the past few years in a particular subfield of AI called machine learning. But what exactly is that? The Economist explains how machine learning works
http://econ.st/1IA3T6V

14:

NOW, Everyone needs to understand it.

to the group: DataScience SG.

Bloomberg Launches "Foundations of Machine Learning" Course for Those with Strong Math Backgrounds

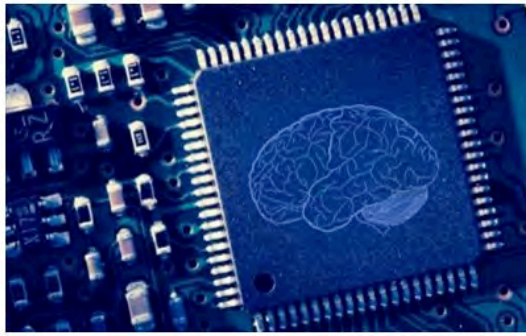


BLOOMBERG.GITHUB.IO

Foundations of Machine Learning

Futurism · Like Page
July 20 at 3:30am ·

Machines can now teach themselves without human interventio
how it works—and how it will transform our society.



Understanding Machine Learning [INFOGRAPH

FUTURISM.COM | BY TODD JAQUITH

Popular Science · Like
July 8 at 3:15am ·

Achieve the impossible by mastering artificial intelligence

Launch Into the Innovative Field of Machine Lear

10 Courses & 63.5 Hours of Training

SHOP.POPSCI.COM

It is far reaching and transformative, disruptive.

Forbes
October 25, 2015 · Like Page · Watch Video

Machine learning started as far back as the 1950s, when computer scientists figured out how to teach a computer to play checkers.

5 Ways Machine Learning Is Reshaping Our World

Machine learning is really just the very advanced application of statistics to learning to identify patterns in data and then make predictions from those patterns.

WWW.FORBES.COM I BY BERNARD MARR

Singularity Hub
May 27 · Like Page · Sign Up

"For overwhelmed researchers navigating the growing stack of science literature — the value isn't in having so much new information, but finding relevant insights when they need them."

Machine Learning's Next Trick Will Transform How Research Is Done

Though research is a slow moving and rigid process, one study shows that the rate of scientific study has exploded in the last 50 years. According to the paper,...

SINGULARITYHUB.COM

The Verge
August 6 at 7:30 PM · Like Page

Some of Dota 2's most beloved names got owned by OpenAI bots.

THEVERGE.COM

The OpenAI Dota 2 bots just defeated a team of former pros

And it wasn't even close

Advertising Age
Yesterday at 6:05am · Like Page

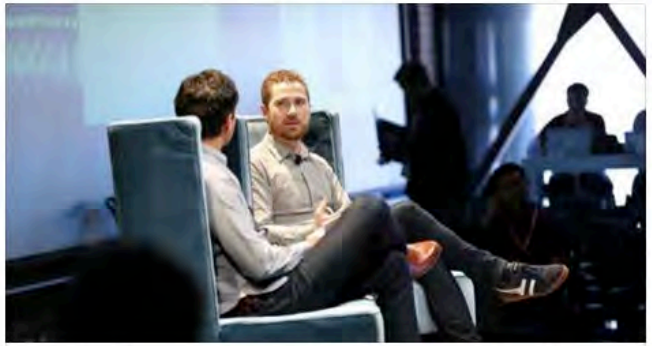
Here are five industries where machine learning will shape customer experience and marketing programs. http://bit.ly/2a7Kosd

When Machines Get Smarter, So Do CMOs

The proliferation of data and intelligent platforms has become increasingly prevalent and is beginning to shape the way businesses must transform their own organizations, writes Evan Rowe.

VentureBeat
July 20 at 9:15pm · Like Page · Message

How machine learning and A.I. are transforming commerce, according to Pinterest



Pinterest's commerce head explains how A.I. and machine learning will transform shopping

Aunkur Arya, Braintree's mobile general manager, and Michael Yamartino, Pinterest's head of commerce, took the stage at VentureBeat's MobileBeat 2016...

calculate how pollution will spread across the city.

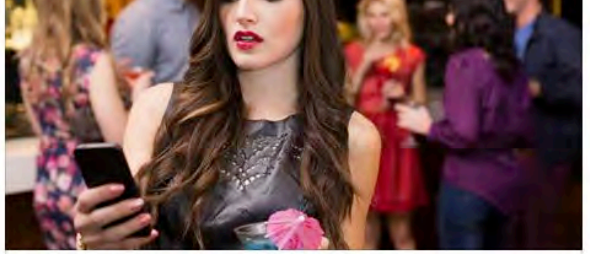
How IBM is trying to use AI to reduce pollution in China

As China's air quality sinks, IBM's research group is using AI techniques to predict pollution levels for every square kilometer 10 days in advance.

TECHNOLOGYREVIEW.COM

How Human-Machine Learning Partnerships Can Reduce Unconscious Bias

ENTREPRENEUR.COM



Machine learning is going to revolutionize the way you use your phone

It'll order you an Uber when you're too drunk, for one.

TECHINSIDER.IO

Digital Trends · July 20 at 8:02am · Like Page · Sign Up

Heading straight to the source.

The key to stopping Ebola? Using machine learning to track infected bats

A team of scientists have used machine learning to create a "bat map" which may help quell the spread of Ebola

DIGITALTRENDS.COM

It can potentially save the world and save us from ourselves

Futurism · April 5 · Like Page · Sign Up

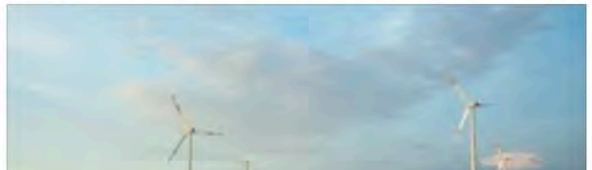
As more and more devices vie for precious bandwidth space, DARPA is looking for creative ways to manage the radio spectrum.

Managing Bandwidth Through Machine Learning

FUTURISM.COM

Scientific American magazine · July 14 at 8:44am · Like Page · Learn More

Although Germany is the world's poster child for renewable energy, its grids cannot yet cope with the erratic nature of wind and solar power.



New Scientist · 11 hrs · Like Page · Sign Up

With US policing in the spotlight, the police are turning to AI to predict which officers are at risk of cracking under stress

NOVA | PBS · July 23 at 3:51am · Like Page · Sign Up

The same techniques that inspired AlphaGo could help Google combat climate change.

Machine Learning Could Dramatically Reduce Google's

Corporates are all-in.

Fortune Magazine · April 14

Like Page · Sign Up

Facebook now uses machine learning to translate 2 billion News Feed items per day.

Inside Facebook's Biggest Artificial Intelligence Project Ever

Facebook's "machine learning" platform is powering your cat videos and family pics.

FOR.TN

http://www.forbes.com/…/how-microsoft-is-making-big-impac…/…

How Microsoft Is Making Big Impact With Machine Learning

During the last two years, Microsoft has upped the ante on Machine Learning

FORBES.COM | BY JANAKIRAM MSV

TechCrunch · February 20

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Salesforce plans to use some of the tech to build out its own machine learning capabilities as part of SalesforceIQ

Salesforce Acquires PredictionIO To Build Up Its Machine Learning Muscle

Salesforce has made another acquisition to build out its technology in machine learning and big data analytics: the company has acquired PredictionIO, a..

TCRN.CH

Business Insider: Tech · July 26 at 5:45am

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Machine learning is a major priority of tech giants like Google, Facebook and Amazon.

Do not ignore machine learning — Google, Facebook, and Amazon are all betting big on it

This is why we can ignore machine learning. It's not about what it can do, but the effects of its prioritization.

BUSINESSINSIDER.COM

Forbes · July 22 at 1:00 PM

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The company is breaking down internal barriers and showing how other companies can do the same.

The Wall Street Journal · May 19

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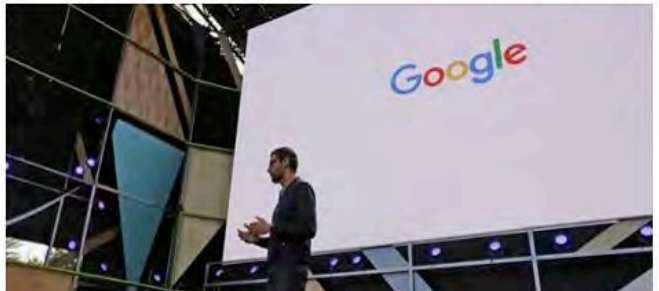
Google is betting it can outsmart rivals on new products—from virtual assistants to messaging apps—with advances in artificial intelligence.



In more ways than one.

Series A funding and acquisitions

NEA - New Enterprise Associates
January 9, 2015 · Like Page · Call Now

GraphLab is now Dato!
Oh, and they've raised $18.5M.
Congratulations!

Machine Learning Startup GraphLab Gets A New Name And An $18.5M Check

Looking to bring machine learning to the masses -- or at least the masses of businesses that could benefit from embedding artificial intelligence in their..

TECHCRUNCH.COM

Two years ago today, I wrote the first lines of code at Starbucks while unemployed and waiting for my UK visa. Happy St Patrick's Day! ☺

WSJ

Inspired by Astrophysics, Machine Learning Startup Wise.io Raises $2.5M - Venture Capital...

The startup's technology analyzes interactions among businesses and their customers.

BLOGS.WSJ.COM | BY DEBORAH GAGE

VentureBeat
July 13 at 3:45pm · Like Page · Message

Self-driving cars may not need pre-loaded detailed 3D maps of the world's roads. This startup promises to map the world in real time as it goes along.

U.K. machine-learning startup FiveAI raises $2.7 million to create the smart autonomous cars of tomo

FiveAI, a U.K. startup leveraging artificial intelligence (A.I.) to build the brains for the autonomous cars of the future, has announced a $2.7 million funding round...

ONVB.CO

I've written exactly, oh, about five machine learning scripts. Anyone want to create a Machine Learning startup in our virtual garage? I'm kidding, almost, but not 100%. The ML world is starting to feel like a mini dot-com 1990's bubble. It's not as easy as putting ".com" on the end of a domain name and writing an ambiguous business plan, not at all, but it has the faint smell of the dot-com bubble.

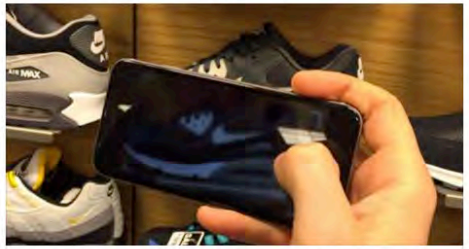
Activity

Record For AI Startups

In Q1'16, there were over 140 deals to startups focused on AI. Khosla Ventures emerges as most active VC investor in AI

VentureBeat
July 6 at 6:49pm · Like Page · Message

Machine vision is advancing, but still early stages.



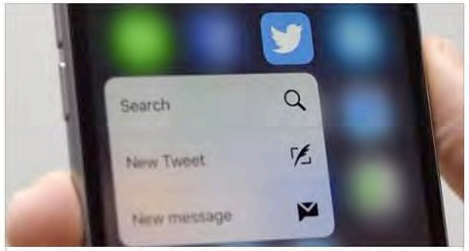
Google buys machine learning startup Moodstock to help your phone's camera identify objects

Google announced today that it has acquired Paris-based Moodstocks, a startup that has developed machine learning technology to bolster the image recognition...

ONVB.CO

The Wall Street Journal
June 21 · Like Page · Use App

Magic Pony's technology allows for better-quality video compression, essentially allowing for better-looking videos that use less data and are less choppy.

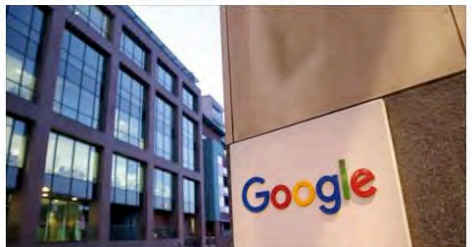


Twitter to Buy Machine-Learning Company Magic Pony

Twitter said Monday that it bought Magic Pony Technology, a London-based machine-learning company, in a move to help bolster its live and streaming video content.

TechCrunch
July 6 at 11:53pm · Like Page · Sign Up

Google is next to acquire a machine learning startup



You might be able to use it today…

TechCrunch
April 10, 2015 ·

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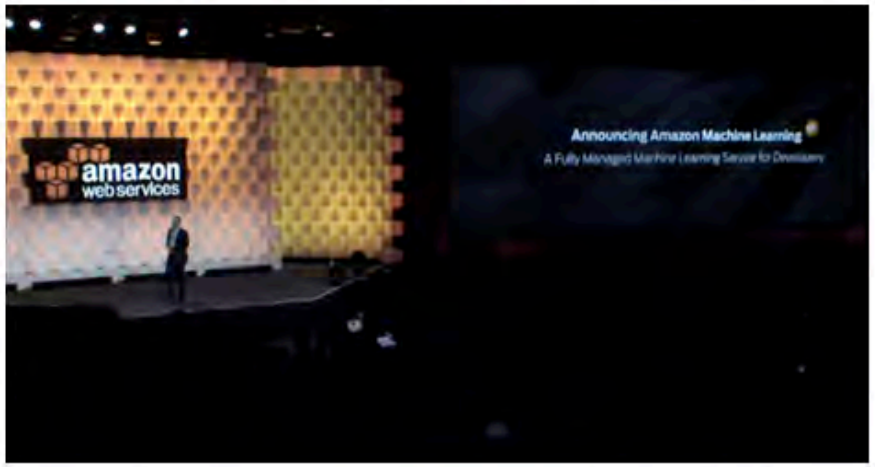
AWS announced a new Machine Learning platform today that it says puts machine learning into the hands of ordinary developers.



AWS Wants To Put Machine Learning In Reach Of Any Developer

Andy Jassy, senior vice president at Amazon Web Services, announced a new machine learning platform today at the AWS Summit in San Francisco. The…

TCRN.CH

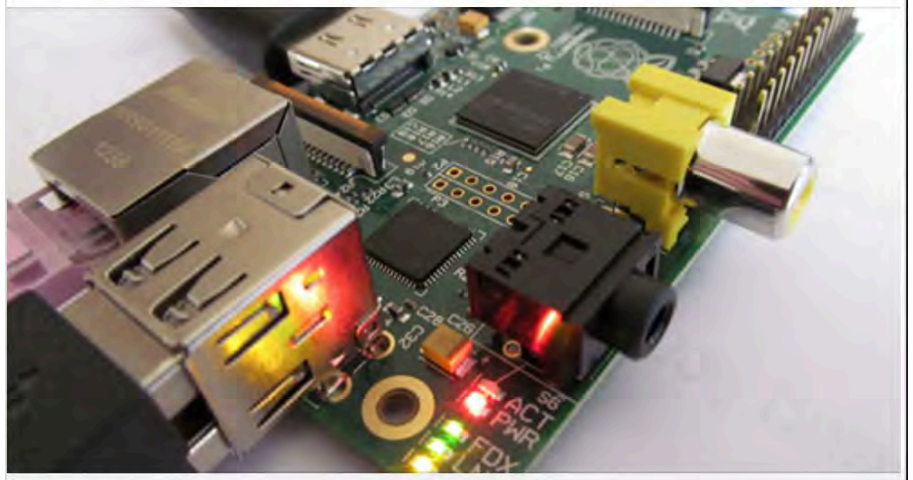
PyTorch Taiwan.

PyTorch…

red a link to the group: Data Artificial Intelligence.

August 5 at 1:32 AM ·

When TensorFlow was first launched in 2015, we wanted it to be an "open source machine learning framework for everyone". To do that, we need to run on as many of the platforms that people are using as possible. We've long supported Linux, MacOS, Windows, iOS, and Android, but despite the heroic efforts of many contributors, running TensorFlow on a Raspberry Pi has involved a lot of work. Thanks to a collaboration with the Raspberry Pi Foundation, we're now happy to say that the latest 1.9 release of TensorFlow can be installed from pre-built binaries using Python's pip package system! If you're running Raspbian 9 (stretch), you can install it by running these two commands from a terminal:



MEDIUM.COM

TensorFlow 1.9 Officially Supports the Raspberry Pi

By Pete Warden, Software Engineer

… but perhaps only a few understand it.

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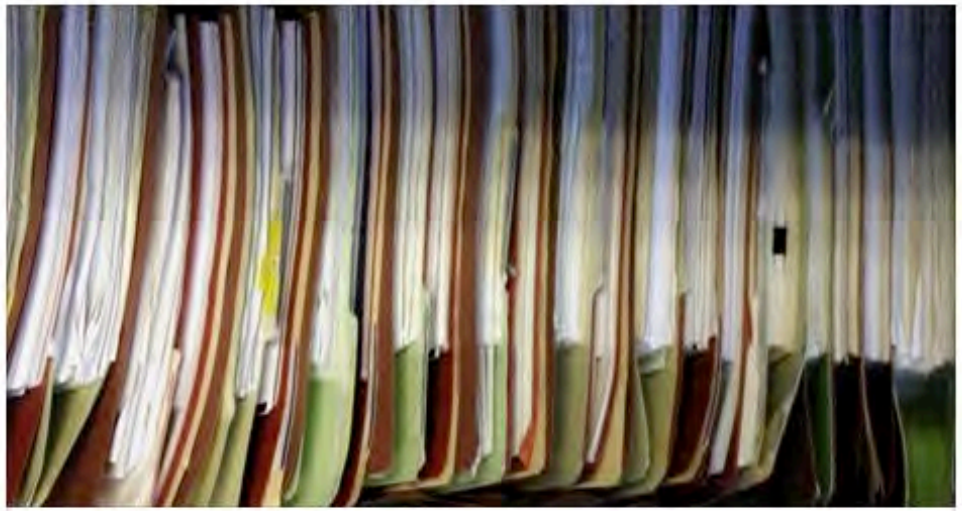
"Machine learning will likely advance in a punctuated equilibrium fashion: in a few, hard-won, big leaps. The reason for this is deep learning software is nearly impossible to debug because we don't fully understand how it works."



The Business Implications of Machine Learning

It's not about what it can do, but the effects of its prioritization

MEDIUM.COM | BY DREW BREUNIG

WIRED

December 13, 2015 ·

Learn More

It is the guiding principle of many applied mathematicians that if something mathematical works really well, there must be a good underlying mathematical reason for it, and we ought to be able to understand it.

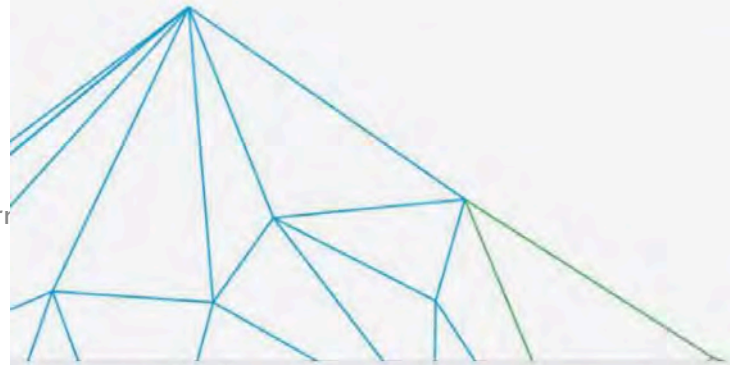
Machine Learning Works Great—Mathematicians Just Don't Know Why

Our current mathematical understanding of many techniques that are central to the ongoing big-data revolution is inadequate, at best.

WIRED.COM | BY INGRID DAUBECHIES

link to the group: DataScience SG.

Bloomberg Launches "Foundations of Machine Learning" Course for Those with Strong Math Backgrounds



BLOOMBERG.GITHUB.IO

Outline for the rest of today

Work through the Week 1 notebook by going through our in-class notebook

Access http://www.comp.nus.edu.sg/~cs3244/2010/01.colab.html and save a copy before starting

Watch the remainder of videos for this week:

- Project Videos (for reference later)
- Administrivia Videos